

RUSTAM LATYPOV

Sublinear Algorithms Group

Aalto University, Finland

I'm a postdoc in the group of Jara Uitto at Aalto University with research and teaching experience in algorithm design and combinatorics. My research interests lie in the mathematical foundations of distributed and parallel computing. I specialize in fundamental graph problems in modern models of computing. During my free time I climb rocks.

rustamlatypov.github.io

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EDUCATION

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|--|--|
| + Postdoctoral researcher • Computer Science
Sublinear Algorithms Group | May 2026 –
<i>Aalto University</i> |
| + Doctoral Researcher • Computer Science
Sublinear Algorithms Group
Supervised by Prof. Jara Uitto | Jun 2021 – April 2026
<i>Aalto University</i>
phd-thesis |
| + Master of Science • Mathematics and Operations Research • GPA 4.9/5
Major in Applied Mathematics • 5/5
Minor in Computer Science • 4.8/5 | 2019 – 2021
<i>Aalto University</i>
rustamlatypov/masters-thesis |
| + Bachelor of Science • Engineering Physics and Mathematics • GPA 4.9/5
Major in Mathematics and Systems Analysis • 5/5
Minor in Computer Science • 5/5 | 2016 – 2019
<i>Aalto University</i>
rustamlatypov/bachelors-thesis |

AWARDS

- ★ Awarded 5000€ – Encouragement grant, The Finnish Foundation for Technology Promotion [\[link\]](#)
- ★ Awarded 5000€ – Nokia Scholarship, Nokia Foundation [\[link\]](#)
- ★ Received 24-month funding for my doctoral studies (out of 20 applicants) – Dept. of CS, Aalto [\[link\]](#)
- ★ Awarded 1000€ for academic success in mathematics – Professor E. J. Nyström Fund [\[link\]](#)
- ★ Awarded $3 \times 500\text{€}$ for general academic success – School of Science, Aalto [\[link\]](#)

PUBLICATIONS (authors are ordered alphabetically, as is standard in the field)

12. Massively parallel algorithms for sparse graphs [\[isbn\]](#)
Rustam Latypov
Aalto University publication series Doctoral Theses 82/2026
11. Near-Optimal Distributed 2-Ruling Sets on Graphs with Low Arboricity [\[arXiv\]](#)
Malte Baumecker, Rustam Latypov, Yannic Maus, Jara Uitto
CoRR 2026.

10. Near-Optimal Directed Low-Diameter Decompositions [\[doi, arXiv\]](#)
Karl Bringmann, Nick Fischer, Bernhard Haeupler, Rustam Latypov
 International Colloquium on Automata, Languages, and Programming (ICALP) 2025
9. A Simple Parallel Algorithm with Near-Linear Work for Negative-Weight
 Single-Source Shortest Paths [\[doi, arXiv\]](#)
Nick Fischer, Bernhard Haeupler, Rustam Latypov, Antti Roeykskoe, Aurelio Sulser
 SIAM Symposium on Simplicity in Algorithms (SOSA) 2025
8. Adaptive Massively Parallel Coloring in Sparse Graphs [\[doi, arXiv\]](#)
Rustam Latypov, Yannic Maus, Shreyas Pai, Jara Uitto
 ACM Symposium on Principles of Distributed Computing (PODC) 2024
7. Conditionally Optimal Parallel Coloring of Forests [\[doi, arXiv\]](#)
Christoph Grunau, Rustam Latypov, Yannic Maus, Shreyas Pai, Jara Uitto
 International Symposium on Distributed Computing (DISC) 2023
6. Adaptive Massively Parallel Connectivity in Optimal Space [\[video, doi, arXiv\]](#)
Rustam Latypov, Jakub Lacki, Yannic Maus, Jara Uitto
 ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2023
5. Fast Dynamic Programming in Trees in the MPC Model [\[doi, arXiv\]](#)
*Chetan Gupta, Rustam Latypov, Yannic Maus, Shreyas Pai, Simo Särkkä,
 Jan Studený, Jukka Suomela, Jara Uitto, Hossein Vahidi*
 ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2023
4. Optimal Deterministic Massively Parallel Connectivity on Forests [\[doi, arXiv\]](#)
Alkida Balliu, Rustam Latypov, Yannic Maus, Dennis Olivetti, Jara Uitto
 ACM-SIAM Symposium on Discrete Algorithms (SODA) 2023
3. Exponential Speedup Over Locality in MPC with Optimal Memory [\[journal, doi, arXiv\]](#)
*Alkida Balliu, Sebastian Brandt, Manuela Fischer, Rustam Latypov,
 Yannic Maus, Dennis Olivetti, Jara Uitto*
 International Symposium on Distributed Computing (DISC) 2022
2. Memory Efficient Massively Parallel Algorithms for LCL Problems on Trees [\[video, doi, arXiv\]](#)
Sebastian Brandt, Rustam Latypov, Jara Uitto
 International Symposium on Distributed Computing (DISC) 2021
1. Coloring Trees in Massively Parallel Computation [\[arXiv\]](#)
Rustam Latypov, Jara Uitto
 CoRR 2021.

ACADEMIC ACTIVITY

Reviewed for Conferences: FOCS'26, PODC'26, STOC'26, SODA'26, FOCS'25, PODC'25, DISC'24, ICALP'24, ITCS'24, SWAT'24, PODC'23, DISC'23, PODC'22, DISC'22, DISC'21, OPODIS'20;
 and Journals: TCS

Teaching at Aalto University

- ◇ Advanced Course in Algorithms – Fall 2022, 2023 (Head TA) [\[link\]](#)
- ◇ Principles of Algorithmic Techniques – Fall 2021, 2023, 2025 (TA / Head TA) [\[link\]](#)

Supervised an MSc thesis

- ◇ Hannes Sederholm, 2022 – *Distributed Drawing of Planar Graphs in the CONGEST model* [\[link\]](#)

PROGRAMMING PROJECTS – github.com/rustamlatypov

- * **Parallel matrix multiplication** C++
Parallel (CPU) matrix multiplication achieving 500-fold speedup w.r.t. sequential
- * **Tile-matching game** C++
Tournament grade Tetris and Pentis with controls in accordance with the Super Rotation System
- * **Parallel radix sort** Scala
Sequential and parallel (CPU) LSD radix sorts achieving 5- and 15-fold speedups w.r.t. scala.quickSort
- * **Inverse problem for resistor networks** MATLAB
Solving non-linear, ill-posed inverse problems for resistor networks both symbolically and numerically
- * **Machine learning classifier for music genres** Python
Solving a skewed, multiclass music genre classification problem with supervised PCA and SVM
- * **Parallel password cracker** Python
A command-line tool for cracking passwords in parallel (CPU) using dictionary and hybrid attacks

WORK EXPERIENCE

- **Postdoctoral researcher** · Aalto University May 2026 –
- **Doctoral researcher** · Aalto University Jun 2021 – April 2026
Worked on the mathematical foundations of distributed and parallel computing. My work mainly focused on graph coloring and connectivity problems in different parallel models of computation.
- **Research assistant** · Aalto University Mar 2020 – May 2021
Worked on distributed graph algorithms in the Massively Parallel Computation model. Developed and published a state-of-the-art graph coloring algorithm (Master's thesis).
- **Giant Leap Intern** · Vaisala Oyj Jun – Aug 2019
Developed software that successfully forecasted test failures in large scale radiosonde production using data mining, feature engineering and machine learning (XGBoost, Python).
- **Research assistant** · Aalto University Jun – Aug 2018
Studied ill-posed non-linear inverse problems for resistor networks both symbolically and numerically (Bachelor's thesis).